Trigonometric Functions

Starter — Prerequisites

Solution 2

1. Sketch $y = \sin x$ for $-180° \leq x \leq 180°$. Use it to solve:

a. $\sin x = 0.8$

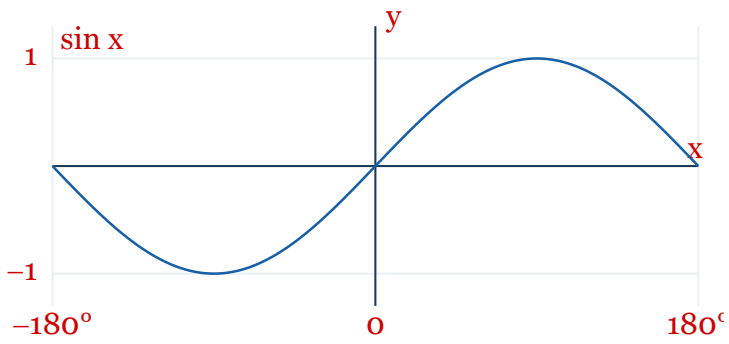
b. $\sin x = -0.4$.

a

a. $x = 53.1°, 126.9°$.

b. $x = -23.6°, -156.4°$ (1 d.p.).

1 sketch



2. Prove $\frac{1}{\sin x \cos x} - \frac{1}{\tan x} = \tan x$.

2

$$\frac{1}{\sin x \cos x} - \frac{\cos x}{\sin x} = \frac{1 - \cos^2 x}{\sin x \cos x}$$
$$= \frac{\sin^2 x}{\sin x \cos x} = \frac{\sin x}{\cos x} = \tan x.$$

3. Solve $3\sin^2(2x) = 1, 0 \leq x \leq 2\pi$.

3

$x = 0.308, 1.26, 1.88, 2.83, 3.45, 4.40, 5.02, 5.98$ (3 s.f.).

Chapter Overview

Solution 3

This chapter is very similar to the trigonometry chapters in Year 1. The only difference is that new trig functions: sec, cosec and cot, are introduced.

1. Reciprocal functions and their graphs

Draw $y = \operatorname{cosec} x$ for $0 \leq x < 2\pi$.

2. Solving

Solve $2\operatorname{cosec}^2 x + \cot x = 5$, $0 \leq x < 2\pi$, to 3 s.f.

3. Proving

Prove $\sec x - \cos x \equiv \sin x \tan x$.

4. Inverse functions: domains and ranges

Understand and sketch arcsin, arccos and arctan.

Small-angle connection

Show that, for small θ,

$\sin 5\theta + \tan 2\theta - \cos 2\theta \approx 2\theta^2 + 7\theta - 1$.

Here θ is in radians.

A new member of the trig family...

Solution 4

$\cos x$

You already know sine, cosine and tangent.

$\cos^2 x = (\cos x)^2$

The latter notation is useful in differentiation (Chapter 9).

$\cos^{-1} x = \arccos x$

Here -1 denotes the inverse function, not a reciprocal power. This historical notation differs from $\cos^2 x$.

$\sec x = \dfrac{1}{\cos x}$

We have convenient names for reciprocals of trigonometric functions.

Reciprocal Trigonometric Functions

Solution 5

$\sec x = \dfrac{1}{\cos x}$

Short for secant: “sehk” or “sea-kant”.

$\operatorname{cosec} x = \dfrac{1}{\sin x}$

Short for cosecant.

$\cot x = \dfrac{\cos x}{\sin x}$

Also $\cot x = 1/\tan x$ wherever both sides are defined.

Cotangent (“cot” rhymes with “pot”).

Tip

Look at the third letter: the c in sec reminds you of 1 over cos.

In proofs, use $\cot x = \cos x/\sin x$.

Sine, Tangent and Secant

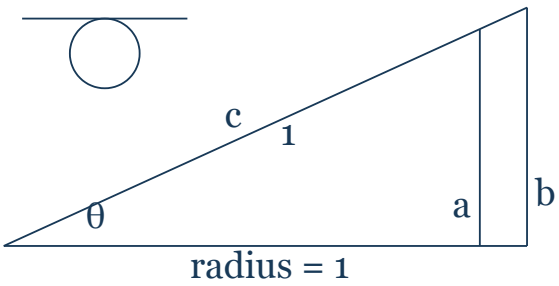
Solution 6

Is ‘tangent’ (tan) in trigonometry related to the ‘tangent’ of a circle?

A tangent is a trigonometric function (which inputs an angle and gives you the ratio between the opposite and adjacent sides of a right-angled triangle), but also a line which touches a circle. Are they related?

$\tan\theta$ = opposite/adjacent.

Just as ‘radius’ can refer either to the line itself or its length, we have names for other special lines, which can also refer to their lengths:



A common myth is that sin, cos and tan are the ‘core’ trigonometric functions.

Actually, they’re sin, tan and *sec*!

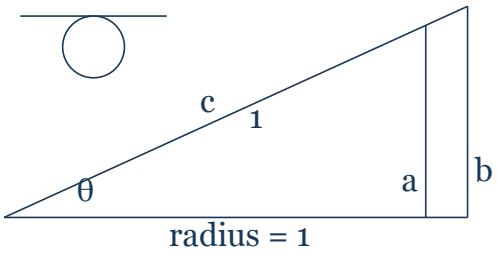
A tangent is a line which touches the circle. We’re interested in the length just between the touching point and where it meets the secant.

Sine (sort of) comes from the word for ‘bowstring’. It refers to half the line if we doubled up the arc and connected the two ends.

A secant (shortened to ‘sec’) is a line which cuts the circle. In a trig setting, we’re interested in the length from the circle centre to where it meets the tangent.

Complementary Trigonometric Functions

Solution 7



Suppose we let the radius of the sector be 1. Then if we define $\sin\theta = \frac{opp}{hyp}$ and $\tan\theta = \frac{opp}{adj}$ and $\sec\theta = \frac{hyp}{adj}$ then this conveniently gives us:

$\sin\theta = \frac{a}{1} = a$ $\quad\quad$ $\tan\theta = \frac{b}{1} = b$

$\sec\theta = \frac{c}{1} = c$

i.e. $\sin\theta$, $\tan\theta$ and $\sec\theta$ actually give us the lengths of the sine, tangent and secant respectively.

And hence $\tan\theta$ and the length of the 'tangent' are clearly linked!

"So if sin, tan and sec are the 'core' trigonometric functions, where does *cos* come into the fray?"

"cosine" (cos), "cotangent" (cot) and "cosecant" (cosec) are the 'complementary' trig functions. Complementary angles add to 90°.

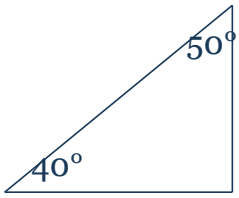
The two smaller angles in a right-angled triangle are clearly complementary.

The complementary sine (cosine) of an angle is by definition the sine of the complementary angle, e.g. $\cos(40°) = \sin(50°)$. Thus:

$\cos(\theta) = \sin(90° - \theta)$ $\quad$ $\operatorname{cosec}(\theta) = \sec(90° - \theta)$

$\cot(\theta) = \tan(90° - \theta)$

The complementary definition gives $\cot\theta = \tan(\pi/2 - \theta) = \cos\theta/\sin\theta$. Interchanging opposite and adjacent leads to $1/\tan\theta$ wherever that reciprocal is defined. The quotient $\cos\theta/\sin\theta$ also defines cotangent where $\cos\theta = 0$.



Calculations

Solution 8

You have a calculator in A Level exams, but it's good however to know how to calculate certain values yourself if needed.

1. $\cot\frac{\pi}{4}$

1
$$\frac{1}{\tan(\pi/4)} = \frac{1}{1} = 1$$

2. $\sec\frac{\pi}{4}$

2
$$\frac{1}{\cos(\pi/4)} = \frac{1}{1/\sqrt{2}} = \sqrt{2}$$

3. $\operatorname{cosec}\frac{\pi}{3}$

3
$$\frac{1}{\sin(\pi/3)} = \frac{2}{\sqrt{3}}$$

4. $\cot\frac{\pi}{6}$

4
$$\sqrt{3}$$

5. $\operatorname{cosec}\frac{5\pi}{6}$

5
$$\operatorname{cosec}\frac{\pi}{6} = 2$$

6. $\cot\frac{\pi}{3}$

6
$$\frac{1}{\sqrt{3}}$$

7. $\sec\frac{\pi}{6}$

7
$$\frac{2}{\sqrt{3}}$$

8. $\operatorname{cosec}\frac{\pi}{2}$

8
$$1$$

9. $\sec\frac{5\pi}{3}$

9
$$\sec\frac{\pi}{3} = 2$$

Exercises 6A, Page 144 — Solution 9

4. Prove $\operatorname{cosec}(\pi - x) = \operatorname{cosec} x$.

4

$$\operatorname{cosec}(\pi - x) = \frac{1}{\sin(\pi - x)} = \frac{1}{\sin x} = \operatorname{cosec} x.$$

5. Show $\cot 30^\circ \sec 30^\circ = 2$.

5

$$\frac{1}{\tan 30^\circ}\frac{1}{\cos 30^\circ} = \frac{1}{1/\sqrt{3}}\frac{1}{\sqrt{3}/2} = 2.$$

6. Write $\operatorname{cosec}\frac{2\pi}{3} + \sec\frac{2\pi}{3} = a + b\sqrt{3}$.

6

$2\pi/3 = \pi - \pi/3$, in quadrant 2.

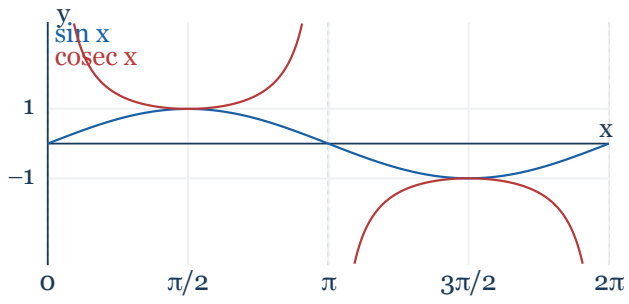
$$\frac{1}{\sin(\pi - \pi/3)} + \frac{1}{\cos(\pi - \pi/3)}$$

$$= \frac{1}{\sin(\pi/3)} - \frac{1}{\cos(\pi/3)} = \frac{2}{\sqrt{3}} - 2$$

$$= -2 + \frac{2}{3}\sqrt{3}.$$

Hence $a = -2,\ b = 2/3$.

Sketches — cosec — Solution 10



Draw $y = \operatorname{cosec} x$, starting with $y = \sin x$.

To draw a graph of $y = \operatorname{cosec} x$, start with a graph of $y = \sin x$, then consider what happens when we reciprocate each y value.

It touches here because the reciprocal of 1 is 1.

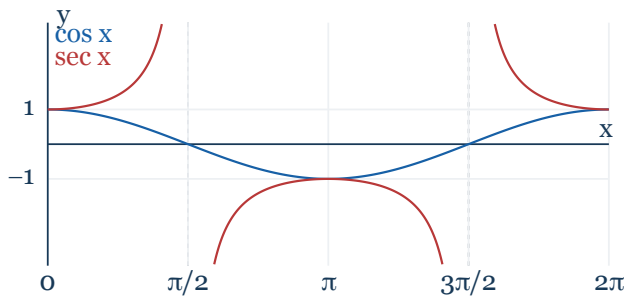
cosec isn't defined for multiples of π because we can't divide by 0. We get an asymptote.

Reciprocating preserves sign. When we divide by a small positive number, we get a very large positive number.

Domain: $x \neq n\pi,\ n \in \mathbb{Z}$.
Range: $y \leq -1$ or $y \geq 1$.

Sketches — sec

Solution 11



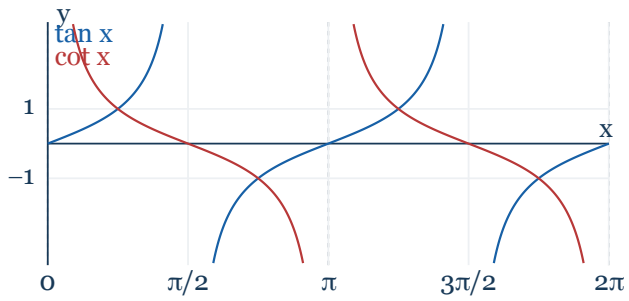
Draw $y = \sec x$, starting with $y = \cos x$.

Domain: $x \neq \frac{\pi}{2} + n\pi,\ n \in \mathbb{Z}$.

Range: $y \leq -1$ or $y \geq 1$.

Sketches — cot

Solution 12



Draw $y = \cot x$, starting with $y = \tan x$.

Extra insight

The relation $\cot\theta = \tan(\pi/2 - \theta)$ reflects the tangent graph in $\theta = \pi/4$. It is defined when $\sin\theta \neq 0$. The input angles θ and $\pi/2 - \theta$ are symmetric about $\pi/4$.

Domain: $x \neq n\pi,\ n \in \mathbb{Z}$.

Range: $y \in \mathbb{R}$.

Example

Solution 13

[Textbook]

a. Sketch the graph of $y = 4\,\text{cosec}\,x, \quad -\pi \leq x \leq \pi$.

b. On the same axes, sketch the line $y = x$.

c. State the number of solutions to the equation $4\,\text{cosec}\,x - x = 0, \quad -\pi \leq x \leq \pi$

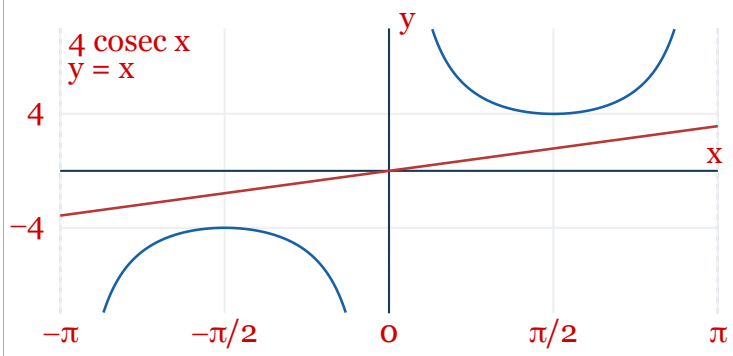
c

$4\,\text{cosec}\,x - x = 0 \iff 4\,\text{cosec}\,x = x.$

The curves do not intersect in the stated interval, so there are no solutions.

Each reciprocal branch satisfies $|4\,\text{cosec}\,x| \geq 4$, while $|x| \leq \pi < 4$.

a–b



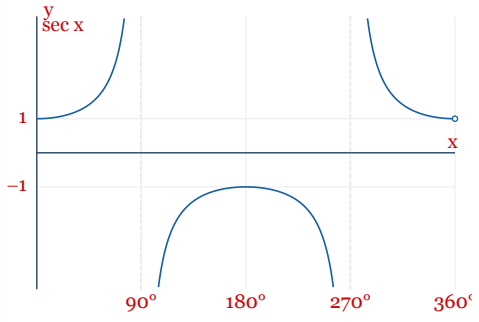
Test Your Understanding

Solution 14

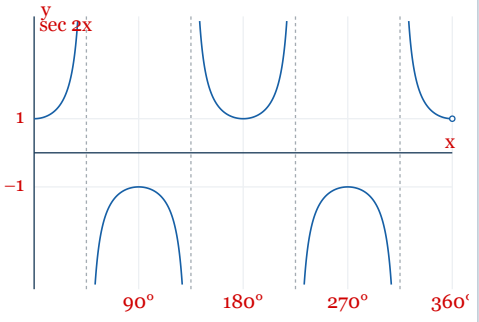
Sketch $y = -1 + \sec 2x$ in the interval $0 \leq x < 360°$.

Draw the transformations stage by stage, unless you feel comfortable doing multiple transformations at once.

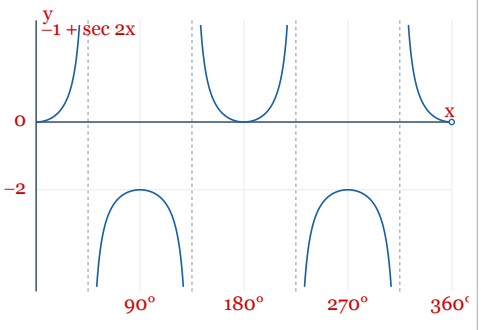
1. Original



2. Compress



3. Translate

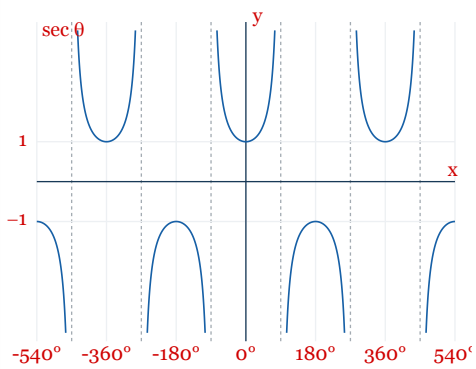


Exercises 6B, Page 148-149 — Solution 15

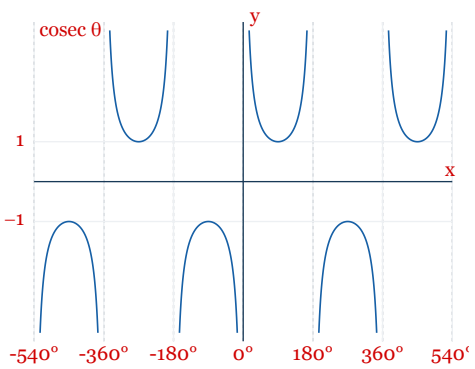
1a. Sketch, for $-540° \leq \theta \leq 540°$:

i. $y = \sec\theta$ ii. $y = \operatorname{cosec}\theta$ iii. $y = \cot\theta$.

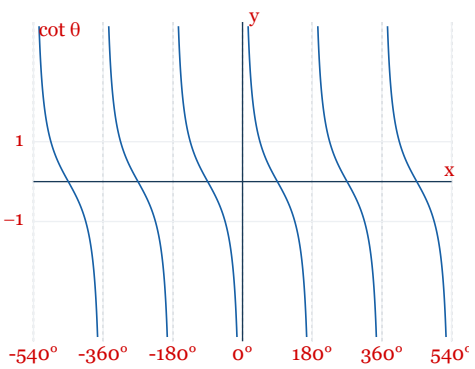
i



ii



iii

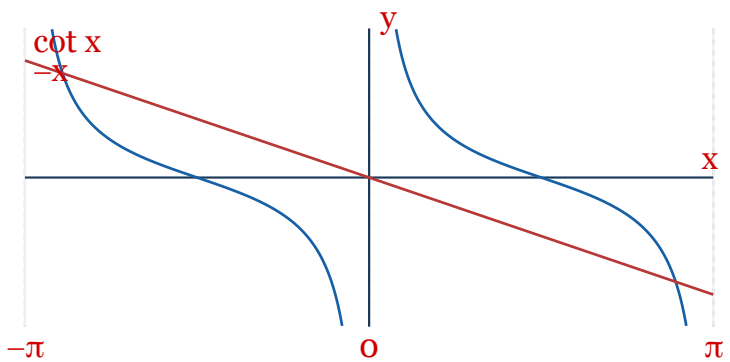


Exercises 6B, Page 148-149 — Solution 16

2a. Sketch $y = \cot x$ and $y = -x$, $-\pi \leq x \leq \pi$, on the same axes.

2b. Deduce the number of solutions of $\cot x + x = 0$ in this interval.

a



b

$\cot x = -x$. There are two intersections, hence two solutions.

Exercises 6B, Page 148-149 — Solution 17

6a. Describe the relationships between these pairs:

i. $\tan(\theta + \pi/2)$, $\tan\theta$.

ii. $\cot(-\theta)$, $\cot\theta$.

iii. $\operatorname{cosec}(\theta + \pi/4)$, $\operatorname{cosec}\theta$.

iv. $\sec(\theta - \pi/4)$, $\sec\theta$.

a

i. Translate $y = \tan\theta$ by $(-\pi/2, 0)$.

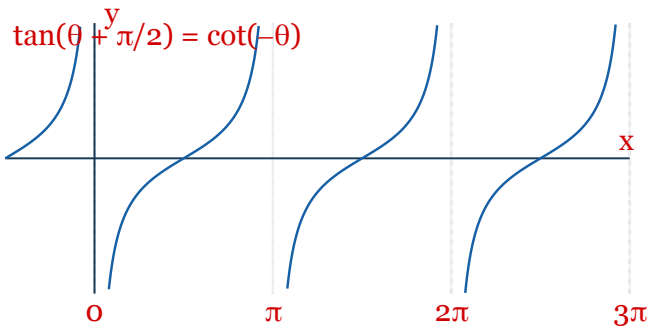
ii. Reflect $y = \cot\theta$ in the y-axis.

iii. Translate $y = \operatorname{cosec}\theta$ by $(-\pi/4, 0)$.

iv. Translate $y = \sec\theta$ by $(\pi/4, 0)$.

6b. Considering the graphs of the four transformed functions, which pairs are equal?

b sketch



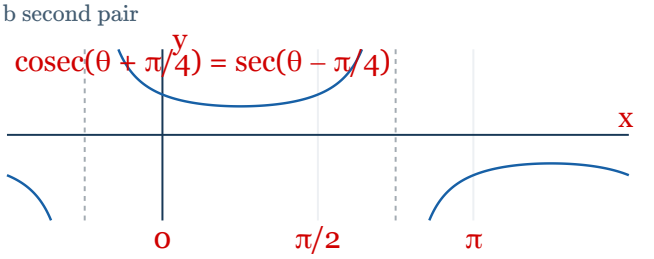
b

$\tan(\theta + \pi/2) = \cot(-\theta) = -\cot\theta$

where defined.

Also $\operatorname{cosec}(\theta + \pi/4) = \sec(\theta - \pi/4)$, since $\sin(\theta + \pi/4) = \cos(\theta - \pi/4)$.

b second pair

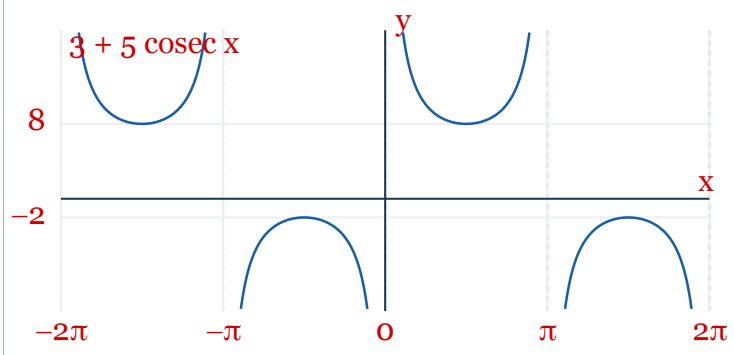


Exercises 6B, Page 148-149 — Solution 18

9a. Sketch $y = 3 + 5\operatorname{cosec} x$, $-2\pi \le x \le 2\pi$.

a Stretch $y = \operatorname{cosec} x$ by scale factor 5 in the y-direction, then translate by $(0, 3)$.

a sketch



9b. Deduce the values of k for which $3 + 5\operatorname{cosec} x = k$ has no solutions.

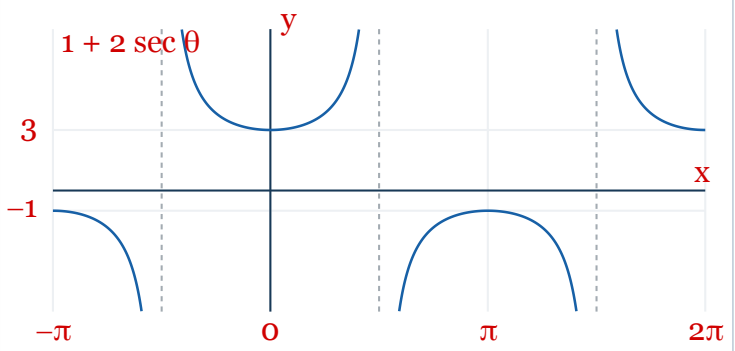
b

The range is $y \le -2$ or $y \ge 8$. There are no solutions when $-2 < k < 8$.

Exercises 6B, Page 148-149 — Solution 19

10a. Sketch $y = 1 + 2\sec\theta$, $-\pi \le \theta \le 2\pi$.

a



10b. State where the gradient is zero.

b

$\theta = -\pi, 0, \pi, 2\pi$, at the maxima and minima.

10c. Deduce the maximum and minimum of $1/(1 + 2\sec\theta)$ and the smallest positive θ at which each occurs.

c

On the negative branch $1 + 2\sec\theta \le -1$, so its reciprocal is at least -1. Minimum: -1, first at $\theta = \pi$.

On the positive branch $1 + 2\sec\theta \ge 3$. Maximum reciprocal: $1/3$, first at $\theta = 2\pi$.

Using sec, cosec and cot — Solution 20

Questions in the exam usually come in two flavours: (a) 'provey' questions requiring to prove some identity and (b) 'solvey' questions.

Tip 1: Get everything in terms of sin and cos first (using $\cot x = \frac{\cos x}{\sin x}$ rather than $\cot x = \frac{1}{\tan x}$)

Tip 2: Whenever you have algebraic fractions being added/subtracted, whether $\frac{a}{b} + \frac{c}{d}$ or $\frac{a}{b} + c$, combine them into one (as we can typically then use $\sin^2 x + \cos^2 x = 1$)

a. Simplify $\sin\theta\cot\theta\sec\theta$.

a

$$\sin\theta\frac{\cos\theta}{\sin\theta}\frac{1}{\cos\theta} = 1.$$

b. Simplify $\sin\theta\cos\theta(\sec\theta + \operatorname{cosec}\theta)$.

b

$$\sin\theta\cos\theta\left(\frac{1}{\cos\theta} + \frac{1}{\sin\theta}\right)$$
$$= \sin\theta\cos\theta\frac{\sin\theta + \cos\theta}{\sin\theta\cos\theta} = \sin\theta + \cos\theta.$$

c. Prove $\frac{\cot\theta\operatorname{cosec}\theta}{\sec^2\theta + \operatorname{cosec}^2\theta} \equiv \cos^3\theta$.

c

$$\frac{\frac{\cos\theta}{\sin\theta}\frac{1}{\sin\theta}}{\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}} = \frac{\cos\theta/\sin^2\theta}{(\sin^2\theta + \cos^2\theta)/(\sin^2\theta\cos^2\theta)}$$
$$= \frac{\cos\theta/\sin^2\theta}{1/(\sin^2\theta\cos^2\theta)} = \cos^3\theta.$$

Multiply numerator and denominator by $\sin^2\theta\cos^2\theta$.

Test Your Understanding

Solution 21

1. Prove $\sec x - \cos x \equiv \sin x \tan x$.

1

$$\text{LHS} = \frac{1}{\cos x} - \cos x = \frac{1 - \cos^2 x}{\cos x}$$
$$= \frac{\sin^2 x}{\cos x} = \sin x \tan x.$$

2. Prove $(1 + \cos x)(\operatorname{cosec} x - \cot x) \equiv \sin x$.

2

$$\text{LHS} = \operatorname{cosec} x - \cot x + \cos x \operatorname{cosec} x - \cos x \cot x$$
$$= \frac{1}{\sin x} - \cot x + \cot x - \frac{\cos^2 x}{\sin x}$$
$$= \frac{1 - \cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

Solving Questions

Solution 22

Solve for $0 \le \theta \le 360°$:

a. $\sec \theta = -2.5$

b. $\cot 2\theta = 0.6$.

a a. $\cos \theta = -0.4$, so $\theta = 113.6°, 246.4°$.

b

b. $0 \le 2\theta \le 720°$, $\tan 2\theta = 5/3$.

$2\theta = 59.0°, 239.0°, 419.0°, 599.0°$.

$\theta = 29.5°, 120°, 210°, 300°$ (3 s.f.).

Solve $\cot \theta = 0$, $0 \le \theta \le 2\pi$.

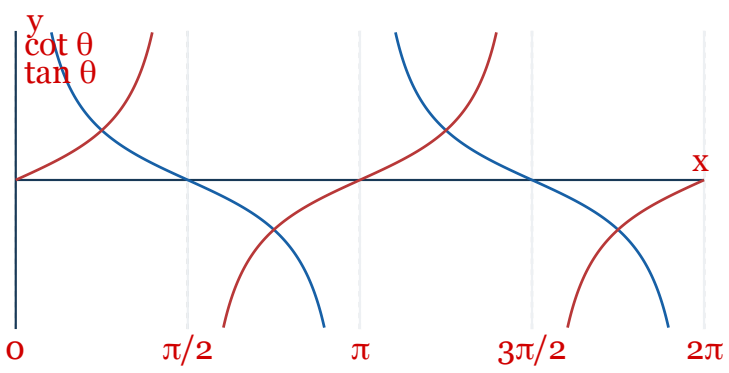
c

Use $\cot \theta = \cos \theta / \sin \theta$. Thus $\cos \theta = 0$, with $\sin \theta \neq 0$:

$\theta = \pi/2,\ 3\pi/2$.

You cannot reciprocate zero. These are the vertical asymptotes of tangent; tangent has no finite value there.

c sketch



Test Your Understanding

Solution 23

Solve in the interval $0 \leq \theta < 360°$:

$\operatorname{cosec} 3\theta = 2$

$0 \leq 3\theta < 1080°$

$\sin 3\theta = \frac{1}{2}$

$3\theta = 30°, 150°, 390°, 510°, 750°, 870°$

$\theta = 10°, 50°, 130°, 170°, 250°, 290°$

Exercises 6C, Page 152

Solution 24

1. Rewrite as powers of $\sec\theta$, $\operatorname{cosec}\theta$ or $\cot\theta$.

a. $\frac{1}{\sin^3\theta}$

a

$\left(\frac{1}{\sin\theta}\right)^3 = \operatorname{cosec}^3\theta$

b. $\frac{4}{\tan^6\theta}$

b

$4\left(\frac{1}{\tan\theta}\right)^6 = 4\cot^6\theta$

c. $\frac{1}{2\cos^2\theta}$

c

$\frac{1}{2}\left(\frac{1}{\cos\theta}\right)^2 = \frac{1}{2}\sec^2\theta$

d. $\frac{1-\sin^2\theta}{\sin^2\theta}$

d

$\frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta$

e. $\frac{\sec\theta}{\cos^4\theta}$

e

$\frac{1}{\cos\theta}\frac{1}{\cos^4\theta} = \frac{1}{\cos^5\theta} = \sec^5\theta$

f. $\sqrt{\operatorname{cosec}^3\theta\cot\theta\sec\theta}$

f

$\sqrt{\frac{1}{\sin^3\theta}\frac{\cos\theta}{\sin\theta}\frac{1}{\cos\theta}} = \sqrt{\frac{1}{\sin^4\theta}} = \operatorname{cosec}^2\theta$

g. $\frac{2}{\sqrt{\tan\theta}}$

g

$2\left(\frac{1}{\tan\theta}\right)^{1/2} = 2\cot^{1/2}\theta$

h. $\frac{\operatorname{cosec}^2\theta\tan^2\theta}{\cos\theta}$

h

$\frac{1}{\sin^2\theta}\frac{\sin^2\theta}{\cos^2\theta}\frac{1}{\cos\theta} = \sec^3\theta$

Exercises 6C, Page 152 — Solution 25

2a. Find $\cot x$ if $5\sin x = 4\cos x$.

a

Divide by $\sin x$:

$$5 = 4\frac{\cos x}{\sin x} \quad \Rightarrow \quad \cot x = \frac{5}{4}.$$

2b. Find $\cot x$ if $\tan x = -2$.

b

$$\cot x = \frac{1}{\tan x} = -\frac{1}{2}.$$

2c. Find $\cot x$ if $\frac{3\sin x}{\cos x} = \frac{\cos x}{\sin x}$.

c

Multiply by $\sin x \cos x$:

$3\sin^2 x = \cos^2 x.$

Divide by $\sin^2 x$:

$$\cot^2 x = 3 \quad \Rightarrow \quad \cot x = \pm\sqrt{3}.$$

Exercises 6C, Page 152 — Solution 26

3. Simplify using the definitions of sec, cosec, cot and tan.

a. $\sin\theta\cot\theta$

a

$$\sin\theta\frac{\cos\theta}{\sin\theta} = \cos\theta$$

b. $\tan\theta\cot\theta$

b

$$\tan\theta\frac{1}{\tan\theta} = 1$$

c. $\tan 2\theta\,\mathrm{cosec}\,2\theta$

c

$$\frac{\sin 2\theta}{\cos 2\theta}\frac{1}{\sin 2\theta} = \frac{1}{\cos 2\theta} = \sec 2\theta$$

d. $\cos\theta\sin\theta(\cot\theta + \tan\theta)$

d

$$\cos\theta\sin\theta\left(\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}\right) = \cos^2\theta + \sin^2\theta = 1$$

e. $\sin^3 x\,\mathrm{cosec}\,x + \cos^3 x\sec x$

e

$$\frac{\sin^3 x}{\sin x} + \frac{\cos^3 x}{\cos x} = \sin^2 x + \cos^2 x = 1$$

f. $\sec A - \sec A\sin^2 A$

f

$$\sec A(1 - \sin^2 A) = \frac{\cos^2 A}{\cos A} = \cos A$$

g. $\sec^2 x\cos^5 x + \cot x\,\mathrm{cosec}\,x\sin^4 x$

g

$$\cos^3 x + \frac{\cos x}{\sin^2 x}\sin^4 x = \cos x(\cos^2 x + \sin^2 x) = \cos x$$

Exercises 6C, Page 153 — Solution 27

7. Solve for $0 \le \theta \le 2\pi$, giving answers in terms of π.

a. $\sec\theta = -1$

b. $\cot\theta = -\sqrt{3}$.

c. $\operatorname{cosec}(\theta/2) = 2\sqrt{3}/3$

d. $\sec\theta = \sqrt{2}\tan\theta,\ \theta \ne \pi/2, 3\pi/2$.

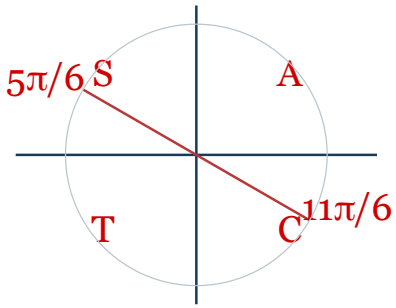
a

a. $\cos\theta = -1$, so $\theta = \pi$.

b

$\tan\theta = -1/\sqrt{3}$. The calculator gives $-\pi/6$, outside the interval. The reference angle is $\pi/6$. Tangent is negative in quadrants 2 and 4: $\theta = \pi - \pi/6,\ 2\pi - \pi/6 = 5\pi/6,\ 11\pi/6$.

b diagram



c

c. $\sin(\theta/2) = 3/(2\sqrt{3}) = \sqrt{3}/2$.

Since $0 \le \theta/2 \le \pi$,

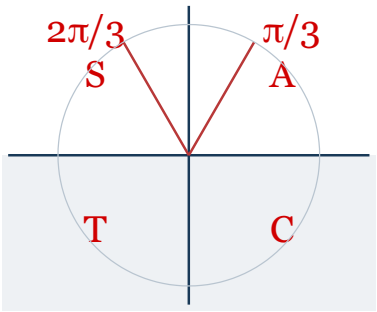
$\theta/2 = \pi/3,\ 2\pi/3 \quad \Rightarrow \quad \theta = 2\pi/3,\ 4\pi/3$.

d

d. $1/\cos\theta = \sqrt{2}\sin\theta/\cos\theta$. Since $\cos\theta \ne 0$,

$\sin\theta = 1/\sqrt{2} \quad \Rightarrow \quad \theta = \pi/4,\ 3\pi/4$.

c diagram

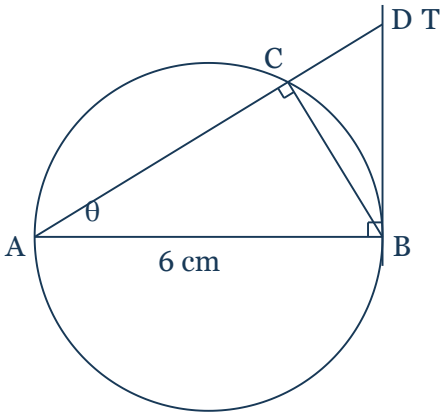


Exercises 6C, Page 153 — Solution 28

8. $AB = 6$ cm is a diameter; BT is tangent at B. Extend chord AC to meet the tangent at D; $\angle DAB = \theta$.

a. Show $CD = 6(\sec\theta - \cos\theta)$ cm.

b. If $CD = 16$ cm, find AC.



a

In right triangle ABD, $AB/AD = \cos\theta$, so $AD = 6\sec\theta$.

In right triangle ACB, $AC/AB = \cos\theta$, so $AC = 6\cos\theta$.

$CD = AD - AC = 6(\sec\theta - \cos\theta)$.

b

$16 = 6\sec\theta - 6\cos\theta \quad \Rightarrow \quad 8 = 3/\cos\theta - 3\cos\theta$.

$8\cos\theta = 3 - 3\cos^2\theta$

$(3\cos\theta - 1)(\cos\theta + 3) = 0$.

$\cos\theta = 1/3$; reject -3. Hence $AC = 6(1/3) = 2$ cm.

Exercises 6C, Page 153 — Solution 29

9a. Prove $\frac{\operatorname{cosec} x - \cot x}{1 - \cos x} = \operatorname{cosec} x$.

a

$$\frac{1/\sin x - \cos x/\sin x}{1 - \cos x}$$
$$= \frac{(1 - \cos x)/\sin x}{1 - \cos x} = \frac{1}{\sin x} = \operatorname{cosec} x.$$

9b. Hence solve $\frac{\operatorname{cosec} x - \cot x}{1 - \cos x} = 2$, $-\pi \leq x \leq \pi$.

b

$$\operatorname{cosec} x = 2 \Rightarrow \frac{1}{\sin x} = 2 \Rightarrow \sin x = \frac{1}{2}.$$

Sine is positive in quadrants 1 and 2:

$x = \pi/6,\ 5\pi/6.$

Exercises 6C, Page 153 — Solution 30

10a. Prove $\frac{\sin x \tan x}{1 - \cos x} - 1 = \sec x$.

a

$$\frac{\sin^2 x}{\cos x(1 - \cos x)} - 1 = \frac{\sin^2 x - \cos x + \cos^2 x}{\cos x(1 - \cos x)}$$
$$= \frac{1 - \cos x}{\cos x(1 - \cos x)} = \frac{1}{\cos x} = \sec x.$$

10b. Explain why $\frac{\sin x \tan x}{1 - \cos x} - 1 = -\frac{1}{2}$ has no solutions.

b

This would require $\sec x = -1/2$, hence $\cos x = -2$, which is impossible for real x.

Exercises 6C, Page 153 — Solution 31

11. Solve $\frac{1+\cot x}{1+\tan x} = 5$, $0 \leq x < 360°$.

11

$$\frac{1+\cos x/\sin x}{1+\sin x/\cos x} = 5$$

$$\frac{(\sin x+\cos x)/\sin x}{(\cos x+\sin x)/\cos x} = 5$$

$$\frac{\sin x+\cos x}{\sin x}\,\frac{\cos x}{\cos x+\sin x} = 5$$

$$\cot x = 5 \quad \Rightarrow \quad \tan x = \frac{1}{5}.$$

The calculator angle is $11.3°$ (1 d.p.). Tangent is positive in quadrants 1 and 3:

$x = 11.3°, 191.3°$.

Both satisfy the original domain restrictions.

New Identities — Solution 32

From Year 1: $\sin^2 x + \cos^2 x = 1$.

Divide by $\cos^2 x$:

$1 + \tan^2 x = \sec^2 x$.

Divide by $\sin^2 x$:

$1 + \cot^2 x = \operatorname{cosec}^2 x$.

Tip

"One is tanned": 1 goes with $\tan^2 x$. Add "co" to both trig functions for the second identity.

Prove $1 + \tan^2 x \equiv \sec^2 x$.

Proof

$$\sin^2 x + \cos^2 x \equiv 1$$

$$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} \equiv \frac{1}{\cos^2 x}$$

$$\tan^2 x + 1 \equiv \sec^2 x.$$

Tip

This has been asked in an exam. Explicitly show each term divided by $\cos^2 x$.

Examples

Solution 33

Prove $\operatorname{cosec}^4\theta - \cot^4\theta = \frac{1+\cos^2\theta}{1-\cos^2\theta}$.

$$\text{LHS} = (\operatorname{cosec}^2\theta + \cot^2\theta)(\operatorname{cosec}^2\theta - \cot^2\theta)$$
$$= \operatorname{cosec}^2\theta + \cot^2\theta = \frac{1}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta}$$
$$= \frac{1+\cos^2\theta}{1-\cos^2\theta} = \text{RHS}.$$

Solve $4\operatorname{cosec}^2\theta - 9 = \cot\theta$, $0 \le \theta \le 360°$.

$$4(1+\cot^2\theta) - 9 = \cot\theta$$
$$4\cot^2\theta - \cot\theta - 5 = 0$$
$$(4\cot\theta - 5)(\cot\theta + 1) = 0.$$
$$\cot\theta = 5/4 \text{ or } -1, \text{ so } \tan\theta = 4/5 \text{ or } -1.$$
$$\theta = 38.7°, 135°, 218.7°, 315°.$$

This is just like in C2; if you had say a mixture of $\sin\theta, \sin^2\theta, \cos^2\theta$: you'd change the $\cos^2\theta$ to $1 - \sin^2\theta$ in order to get a quadratic in terms of sin.

Test Your Understanding

Solution 34

Edexcel C3 June 2013 (R), 6(ii). Solve $3\sec^2\theta + 3\sec\theta = 2\tan^2\theta$, $0 \le \theta < 2\pi$. Show all working; give exact answers.

$$3\sec^2\theta + 3\sec\theta = 2(\sec^2\theta - 1)$$
$$\sec^2\theta + 3\sec\theta + 2 = 0$$
$$(\sec\theta + 2)(\sec\theta + 1) = 0.$$
$$\cos\theta = -1/2 \Rightarrow \theta = 2\pi/3, 4\pi/3.$$
$$\cos\theta = -1 \Rightarrow \theta = \pi.$$

Solve $2\operatorname{cosec}^2 x + \cot x = 5$, $0 \le x < 2\pi$, to 3 s.f.

2

$$2(1+\cot^2 x) + \cot x - 5 = 0$$
$$2\cot^2 x + \cot x - 3 = 0$$
$$(2\cot x + 3)(\cot x - 1) = 0.$$
$$\cot x = -3/2 \text{ or } 1, \text{ so } \tan x = -2/3 \text{ or } 1.$$
$$x = 0.785, 2.55, 3.93, 5.70 \text{ (3 s.f.)}.$$

Exercises 6D, Pages 156-157 — Solution 35

7. Given $3\tan^2\theta + 4\sec^2\theta = 5$ and θ obtuse, find the exact value of $\sin\theta$.

7

$3\tan^2\theta + 4(1+\tan^2\theta) = 5$

$7\tan^2\theta = 1 \quad \Rightarrow \quad \tan^2\theta = 1/7$

$\cot^2\theta = 7 \quad \Rightarrow \quad \operatorname{cosec}^2\theta - 1 = 7$

$\operatorname{cosec}^2\theta = 8 \quad \Rightarrow \quad \sin^2\theta = 1/8.$

θ is in quadrant 2, where sine is positive:

$$\sin\theta = \sqrt{1/8} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}.$$

Exercises 6D, Pages 156-157 — Solution 36

12. Given $2\sec^2\theta - \tan^2\theta = p$, show that $\operatorname{cosec}^2\theta = \frac{p-1}{p-2}$, $p \neq 2$.

12

$2(1+\tan^2\theta) - \tan^2\theta = p$

$$\tan^2\theta = p - 2 \quad \Rightarrow \quad \cot^2\theta = \frac{1}{p-2}.$$

$$\operatorname{cosec}^2\theta = 1 + \cot^2\theta = 1 + \frac{1}{p-2}$$

$$= \frac{(p-2)+1}{p-2} = \frac{p-1}{p-2}, \qquad p \neq 2.$$

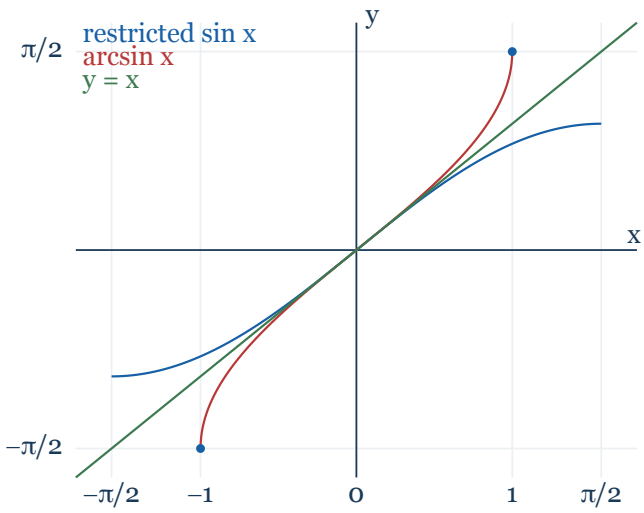
Inverse Trig Functions

Solution 37

You need to know how to sketch
$y = \arcsin x, y = \arccos x, y = \arctan x$.
(Yes, you could be asked in an exam!)

Before taking the inverse of sine, restrict its domain to $-\pi/2 \leq x \leq \pi/2$. Why?
Only one-to-one functions have inverse functions. The restriction makes sine one-to-one.
Reflect its graph in $y = x$.

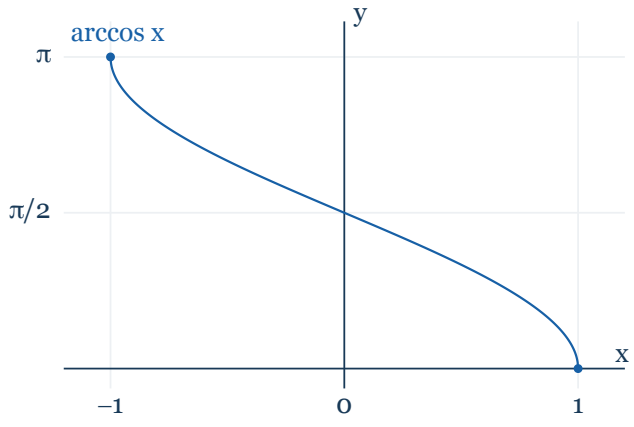
For arcsin: domain $[-1, 1]$, range $[-\pi/2, \pi/2]$.



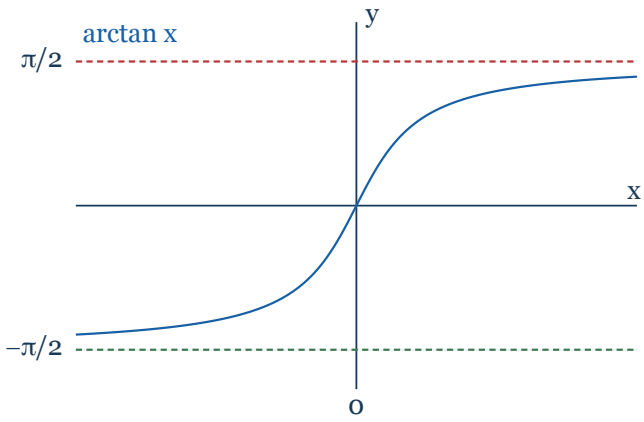
Inverse Trig Functions

Solution 38

$y = \arccos x$
Domain $[-1, 1]$; range $[0, \pi]$.



$y = \arctan x$
Domain $\mathbb{R}$; range $(-\pi/2, \pi/2)$.
Horizontal asymptotes: $y = \pm\pi/2$.



Evaluating inverse trig functions

Solution 39

[Textbook] Work out, in radians, the values of:

$\arcsin\left(-\frac{\sqrt{2}}{2}\right)$

$\arccos(-1)$

$\arctan\left(\sqrt{3}\right)$

You can simply use the $\sin^{-1}x$, $\cos^{-1}x$ and $\tan^{-1}x$ buttons on your calculator.

If you don't have a calculator, just use the sin, cos, tan graphs backwards.

a

$\arcsin(-\sqrt{2}/2) = -\pi/4.$

b

$\arccos(-1) = \pi.$

c

$\arctan(\sqrt{3}) = \pi/3.$

One Final Problem — Edexcel C3 January 2007

Solution 40

Edexcel C3 January 2007, 8(ii).

Given $y = \arccos x$, $-1 \leq x \leq 1$, $0 \leq y \leq \pi$:

a. Express $\arcsin x$ in terms of y.

b. Hence evaluate $\arccos x + \arcsin x$, in terms of π.

Fewer than 10% of candidates got this part right.

a

$x = \cos y = \sin(\pi/2 - y).$

$\pi/2 - y \in [-\pi/2, \pi/2]$, the principal range of arcsin. Thus

$\arcsin x = \pi/2 - y.$

b

$\arccos x + \arcsin x = y + (\pi/2 - y) = \pi/2.$

Exercises 6E, Pages 160-161 — Solution 41

3. Without a calculator, evaluate:

a. $\sin(\arcsin \frac{1}{2})$

b. $\sin(\arcsin(-\frac{1}{2}))$.

c. $\tan(\arctan(-1))$

d. $\cos(\arccos 0)$.

a

a. Let $\arcsin(1/2) = \alpha$, so $\sin\alpha = 1/2$, $-\pi/2 \le \alpha \le \pi/2$. Thus $\alpha = \pi/6$, and $\sin(\pi/6) = 1/2$.

b

b. $\arcsin(-1/2) = -\pi/6$, within the principal range. Thus $\sin(-\pi/6) = -1/2$.

c

c. Let $\arctan(-1) = \alpha$. Then $\tan\alpha = -1$, $-\pi/2 < \alpha < \pi/2$, so $\alpha = -\pi/4$. Therefore the value is -1.

d

d. Let $\arccos 0 = \alpha$. Then $\cos\alpha = 0$, $0 \le \alpha \le \pi$, so $\alpha = \pi/2$. Therefore the value is 0.

Exercises 6E, Pages 160-161 — Solution 42

4a. Evaluate $\sin(\arccos(1/2))$ exactly, without a calculator.

a

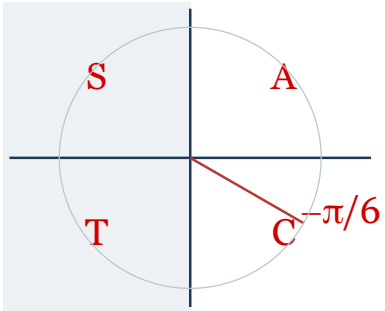
$\arccos(1/2) = \pi/3$. Hence $\sin(\pi/3) = \sqrt{3}/2$.

4b. Evaluate $\cos(\arcsin(-1/2))$.

b

$\arcsin(-1/2) = -\pi/6$, in quadrant 4, where cosine is positive. Hence $\cos(-\pi/6) = \sqrt{3}/2$.

b diagram

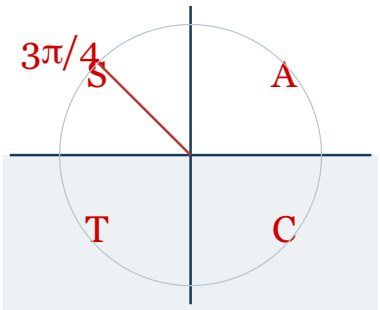


4c. Evaluate $\tan(\arccos(-\sqrt{2}/2))$.

c

Let $\arccos(-\sqrt{2}/2) = \alpha$. Since $0 \le \alpha \le \pi$, the angle is in quadrant 2: $\alpha = 3\pi/4$. Hence $\tan(3\pi/4) = -1$.

c diagram



Exercises 6E, Pages 160-161

Solution 43

5. Given $\arcsin k = \alpha$, $0 < k < 1$, with α in radians, write the first two positive solutions of $\sin x = k$ in terms of α.

5

Because $k > 0$, the first positive solutions are in quadrants 1 and 2:

$x = \arcsin k, \quad \pi - \arcsin k$

$x = \alpha, \quad \pi - \alpha.$

Try a few numerical values of k.

Exercises 6E, Pages 160-161

Solution 44

6. Given $\arcsin x = k$, $0 < k < \pi/2$:

a. State the possible values of x.

b. Express i. $\cos k$, ii. $\tan k$, in terms of x.

c. How do the answers to b change if $-\pi/2 < k < 0$?

a

a. $x = \sin k$. Sine is increasing on the principal interval, so $0 < \sin k < \sin(\pi/2)$, hence $0 < x < 1$.

b(i)

b(i). $\cos k = \pm\sqrt{1 - \sin^2 k} = \pm\sqrt{1 - x^2}$. Since k is in quadrant 1, $\cos k > 0$:

$\cos k = \sqrt{1 - x^2}$.

b(ii)

b(ii) $\tan k = \dfrac{\sin k}{\cos k} = \dfrac{x}{\sqrt{1 - x^2}}$.

c

c. k is now in quadrant 4. Cosine remains positive, so the formula for $\cos k$ is unchanged. Tangent changes sign because $x < 0$; its formula is unchanged.

Exercises 6E, Pages 160-161 — Solution 45

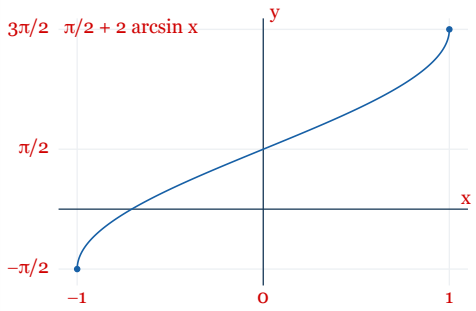
7. Sketch:

a. $y = \pi/2 + 2\arcsin x$ b. $y = \pi - \arctan x$ c. $y = \arccos(2x + 1)$.

a

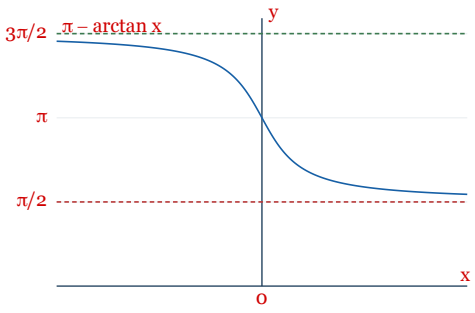
Stretch $y = \arcsin x$ by factor 2 vertically, then translate by $(0, \pi/2)$.

a sketch



b Reflect $y = \arctan x$ in the x-axis, then translate by $(0, \pi)$.

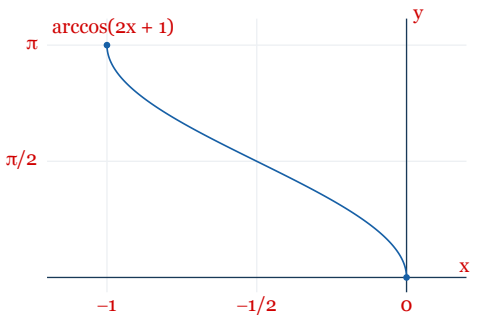
b sketch



c

Translate $y = \arccos x$ by $(-1, 0)$, then stretch horizontally by factor $1/2$.

c sketch

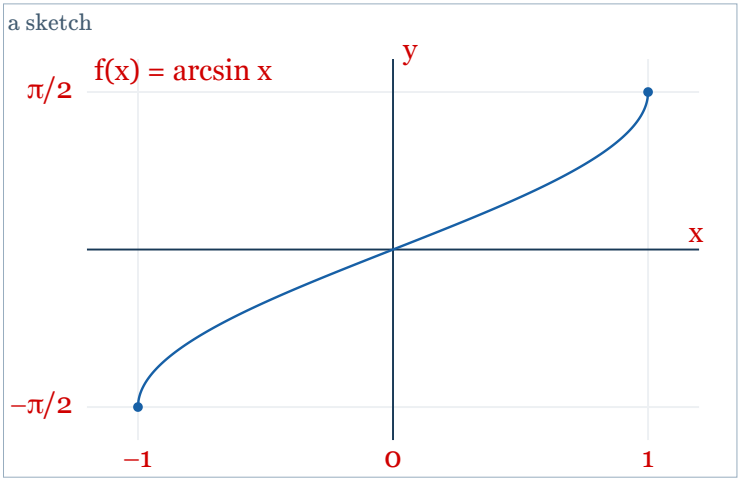


Exercises 6E, Pages 160-161 — Solution 46

8. $f(x) = \arcsin x$, $-1 \le x \le 1$, and $g(x) = f(2x)$.

a. Sketch $y = f(x)$ and state the range of f.

a sketch



a range

$-\pi/2 \le f(x) \le \pi/2$.

b. Sketch $y = g(x)$.

b Compress the graph of $y = f(x)$ horizontally by scale factor $1/2$. The domain is $[-1/2, 1/2]$.

b sketch

